
title: K · Integration Specifications description: API contract, auth model, rate limits, historical outage patterns, retry strategy, and testability surface for each of the nine regulator integrations required by the PBD — GSIS, Pag-IBIG, PhilHealth, SSS, BIR, CSC, DBM, DICT, and PhilSys.

K · Integration Specifications

!!! info “How this paper is used” [Paper C](#) lists the nine external integrations at the container level; [Paper A](#) names them by regulator. This paper is what an engineer opens on day one of M2 to write the first line of integration code — a contract, an auth model, a rate limit, a retry policy, and a testability plan, per regulator.

Contracts and rate limits below reflect ****publicly documented behaviour**** and community-reported

K.1 Contract discipline — what every integration must have

Every regulator integration in this deck ships with **six artefacts** before any production traffic:

1. **OpenAPI 3.1 contract** (or documented protocol equivalent for non-REST).
2. **Auth handshake diagram** — issuer, key rotation cadence, replay protection.
3. **Rate-limit and backoff policy** — request/second ceiling, burst budget, retry ladder.
4. **Idempotency plan** — how a retry does not double-process.
5. **Test fixtures + mock server** — reproducible responses for CI without depending on the sandbox.
6. **Runbook** — what to do when the regulator is down, delayed, or returns malformed data.

If any of the six is missing, the integration is **not** promoted past SIT. This is a design gate, not a wish.

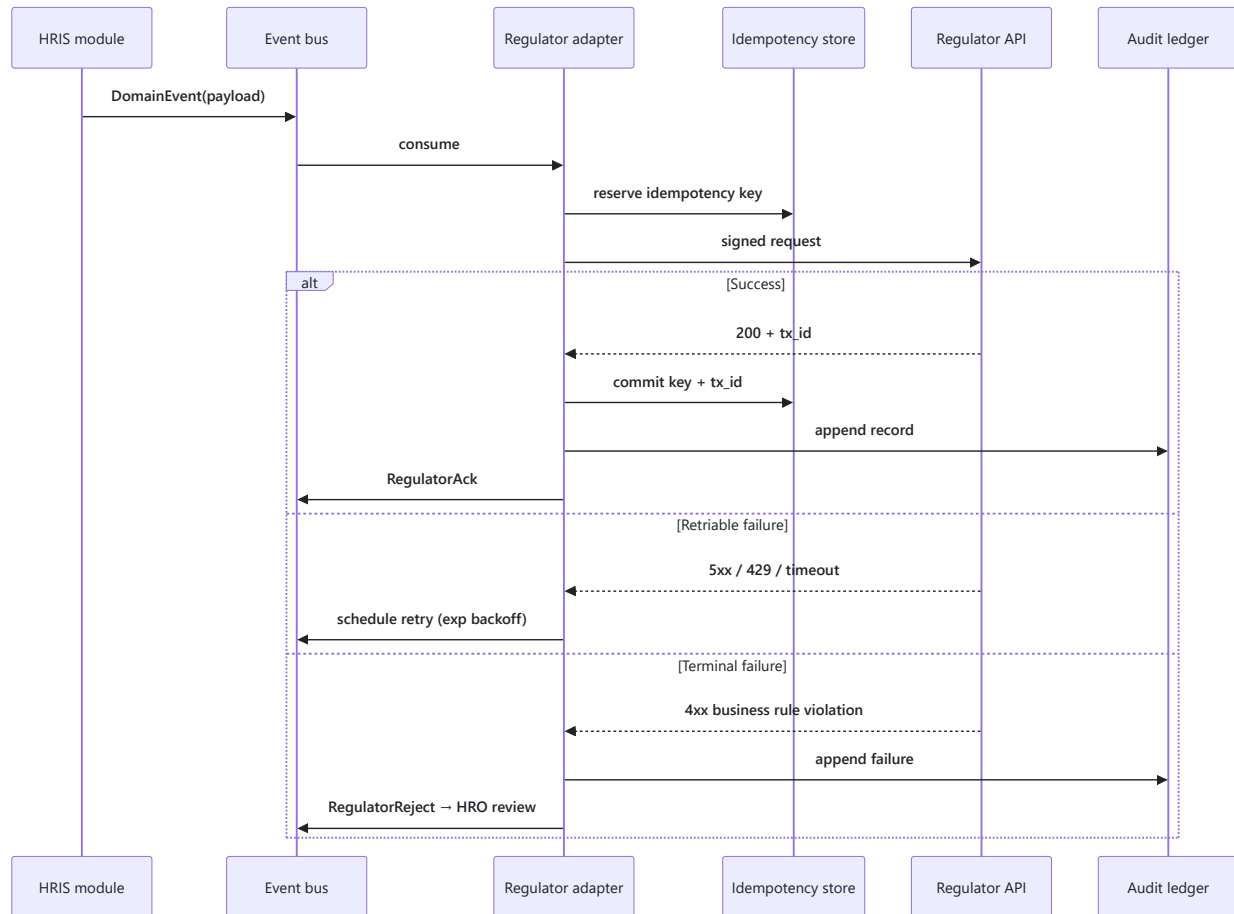
K.2 Regulator inventory — one-line each

#	Regulator	Protocol	Purpose	Est. calls / month	Criticality
1	GSIS	REST + JWS	Loan / retirement remittance	~ 2 M	● Payroll-critical
2	Pag-IBIG	REST	HDMF contribution + loan	~ 2 M	● Payroll-critical
3	PhilHealth	SOAP + REST	Health premium remittance	~ 2 M	● Payroll-critical
4	SSS	REST	SSS premium (COS / JO staff)	~ 200 K	● Payroll-material
5	BIR	REST + eBIRForms	Withholding tax + reporting	~ 2 M + 12 K forms	● Payroll-critical
6	CSC	REST	Position + eligibility + Form 9	~ 400 K	● Recruitment-critical
7	DBM	REST + file drop	PSIPOP position master + WFP	~ 1.1 M ref / mo	● Plantilla-critical
8	DICT	REST + policy	eGov interop + PhilGov cert	Ad-hoc	● Compliance
9	PhilSys	REST + mTLS	eKYC verify + demographic	~ 3 M (peak migration)	● Identity-critical

Non-regulator but architecturally similar: **Landbank** disbursement API (already listed in [Paper C](#)). Not a “regulator” in the RA 10173 sense but shares the integration pattern.

K.3 Common design — everything anchors to this

Before per-regulator detail, every integration inherits the same skeleton:



Non-negotiable properties:

- **Every request is signed** — JWS or mTLS, per regulator; no unsigned production traffic ever.
- **Every request has an idempotency key** — reused across retries; regulators that do not accept an idempotency header have one synthesised in the request payload.
- **Every response is written to the audit ledger** — including hash of request body, response status, and regulator's transaction ID.
- **No adapter has direct access to source-of-truth tables** — it reads from the outbox and writes to inbox tables owned by its bounded context (transactional outbox pattern).
- **Circuit-breaker + bulkhead** — an unhealthy regulator cannot exhaust the thread pool for the healthy ones.

K.4 Per-regulator specifications

K.4.1 GSIS

Aspect	Detail
Auth	JWS signature with rotating RSA keypair; keys re-registered every 12 months
Protocol	REST / JSON
Endpoints used	<code>/members/{gsis_no}</code> , <code>/remittance/monthly</code> , <code>/loan-status/{gsis_no}</code>
Rate limit	~ 60 req/s per client; monthly bulk endpoint accepts 100 K rows/batch
Peak call pattern	Concentrated on payroll cutoff (day 10 and day 25 of month)
Timeout	30 s per call; batch endpoint 5 min
Retry policy	Exponential backoff 2, 4, 8, 16, 32 s; give up after 5; move to next-cycle retry
Idempotency	Header <code>X-GSIS-Idempotency-Key</code> ; regulator dedups for 24 h
Historical outage pattern	Scheduled maintenance nightly 00:00–02:00 PHT; occasional unannounced outages during month-end
Fallback	Queue for next window; anomaly detector flags stalled remittances at 24 h
Data pushed	Employee GSIS number, salary base, contribution, cycle period
Data pulled	Loan balances, retirement eligibility, membership status
Audit anchor	Every push includes GSIS transaction ID stored in <code>payroll.gsis_remit.regulator_tx_id</code>

K.4.2 Pag-IBIG (HDMF)

Aspect	Detail
Auth	API key + IP allow-list + HMAC signature
Protocol	REST / JSON
Endpoints used	<code>/employer/remittance</code> , <code>/member/{mid}/loans</code> , <code>/member/{mid}/contributions</code>
Rate limit	~ 40 req/s; bulk remittance 50 K rows/batch
Peak call pattern	Payroll cutoff + 25 of month
Timeout	30 s
Retry policy	Same 5-step exponential ladder
Idempotency	Payload includes <code>client_ref</code> ; regulator dedups for 48 h
Historical outage pattern	Occasional partial outages when member portal is under load
Fallback	Batch-file drop (SFTP) is a documented alternative; adapter can switch
Special note	HDMF has a separate loan-remittance workflow; do not mix with monthly contribution

K.4.3 PhilHealth

Aspect	Detail
Auth	Digital certificate (PKI) + SOAP-WS-Security header
Protocol	SOAP (legacy) with a REST wrapper for newer endpoints
Endpoints used	SubmitRF1 , RegisterMember , RetrieveContributions
Rate limit	~ 20 req/s per client; RF1 batch accepts up to 10 K rows
Peak call pattern	Month-end RF1 submission
Timeout	45 s (SOAP is slow)
Retry policy	3 retries only; PhilHealth idempotency is fragile — over-retrying causes duplicate contribution records
Idempotency	Not natively supported; adapter enforces via local dedup table + reconciliation report
Historical outage pattern	Weekend maintenance windows; occasional SSL cert rotation issues
Fallback	Manual RF1 upload via portal remains a documented last-resort
Special note	This is the highest-risk regulator integration — plan the retry ceiling and reconciliation report as first-class controls

K.4.4 SSS

Aspect	Detail
Auth	JWT bearer + IP allow-list
Protocol	REST / JSON
Endpoints used	/employer/contribution , /employer/loan-remittance , /member/status
Rate limit	~ 30 req/s
Peak call pattern	Cutoff + 15 for COS / JO staff
Timeout	30 s
Retry policy	Standard 5-step exponential
Idempotency	X-SSS-Ref header
Historical outage pattern	Nightly maintenance; occasional slow response during peak
Data volume	Lower than GSIS (COS / JO only; plantilla staff are on GSIS not SSS)

K.4.5 BIR

Aspect	Detail
Auth	Client certificate + shared secret per SDO branch
Protocol	REST for real-time; eBIRForms XML upload for filings
Endpoints used	Withholding-tax computation validators; eBIRForms Form 1601-C / 1604-CF
Rate limit	~ 25 req/s real-time; eBIRForms limits by portal quota
Peak call pattern	Payroll cutoff + monthly / quarterly / annual filing windows
Timeout	45 s
Retry policy	3 retries + human escalation
Idempotency	Filing IDs; overwrites are audit-visible
Historical outage pattern	Filing-deadline day slowdowns (April 15 and quarterly)
Special note	Form generation is deterministic from payroll ledger; if BIR is down, forms are held in outbox and released when connectivity restores

K.4.6 CSC

Aspect	Detail
Auth	OAuth2 client credentials + JWT
Protocol	REST / JSON
Endpoints used	/positions/{code} , /eligibility/{philsys} , /form9/submit , /personnel-actions
Rate limit	~ 15 req/s
Peak call pattern	Recruitment posting windows + appointment issuance
Timeout	30 s
Retry policy	Standard 5-step
Idempotency	Native — X-CSC-Client-Ref
Historical outage pattern	Predictable maintenance Sundays
Data pushed	CSC Form 9 (job posting), personnel action reports
Data pulled	Eligibility records, position-classification master, active issuances

K.4.7 DBM (PSIPOP)

Aspect	Detail
Auth	mTLS + IP allow-list
Protocol	REST + monthly file drop (CSV over SFTP)
Endpoints used	/psipop/positions , /psipop/vacancies , /wfp/employees
Rate limit	~ 10 req/s live; nightly full-master ~ 1.1 M rows
Peak call pattern	Monthly plantilla reconciliation
Timeout	60 s
Retry policy	For live: 5-step; for file drop: retry with new file next window
Idempotency	Position codes are the natural key; PATCH is idempotent by design
Historical outage pattern	Batch endpoint occasionally slow at end of fiscal year
Special note	DBM PSIPOP is authoritative for plantilla items; conflicts resolve in DBM's favour

K.4.8 DICT

Aspect	Detail
Auth	Per-service — GovCloud PH uses IAM; eGovPH uses OIDC
Protocol	Varies per service
Usage	Certificate authority for code signing + eGovPH SSO handshake (future) + DICT security-advisory feed
Rate limit	Very low; primarily advisory / compliance
Peak call pattern	On advisory publication (aperiodic)
Special note	DICT is more a compliance regulator than a live integration — the HRIS ingests advisories rather than transacting

K.4.9 PhilSys

Aspect	Detail
Auth	mTLS + client certificate (per PSA-issued)
Protocol	REST / JSON
Endpoints used	<code>/verify/{philsys_no}</code> (eKYC), <code>/authenticate</code> (biometric-optional)
Rate limit	~ 50 req/s per client; higher during onboarding waves
Peak call pattern	Migration wave M3 (initial enrolment); ESS first-login
Timeout	20 s
Retry policy	3 retries then move to manual verification queue
Idempotency	X-PhilSys-Ref header
Historical outage pattern	Rare; PSA-run infrastructure
Special note	Used for the D §13 PhilSys eKYC dedup value-add; the migration pipeline in Paper H §H.5 is the biggest single consumer

K.5 Cross-cutting concerns

K.5.1 Rate-limit budgeting

Each adapter carries a rate budget. If two integrations peak concurrently — the payroll cutoff hits GSIS + Pag-IBIG + PhilHealth + SSS + BIR together — the aggregate outbound rate must stay under DepEd’s own egress limits and the regulators’ ceilings.

Time window	Concurrent integrations	Aggregate ceiling	Design headroom
Payroll cutoff (day 10, 25)	GSIS + Pag-IBIG + PhilHealth + SSS + BIR	175 req/s total	40% headroom → 105 req/s target
Month-end filing	BIR + PhilHealth RF1	45 req/s	40% headroom
Recruitment posting	CSC + PhilSys	65 req/s	50% headroom
Migration wave	PhilSys + DBM PSIPOP	60 req/s	50% headroom

Adapter queues use **priority + fair-share** — payroll-critical requests preempt discretionary ones (e.g. eligibility lookups) during contention.

K.5.2 Testability — mock everything, always

Every adapter ships with a **stateful mock server** that speaks the same contract as the regulator sandbox. CI runs full end-to-end tests against the mock. The sandbox is used only for **contract-drift detection** (weekly) and for the pre-M5 dress rehearsal.

Mock servers live in `platform/mocks/{gsis,pagibig,philhealth,sss,bir,csc,dbm,dict,philsys}` and are versioned alongside the adapters.

K.5.3 Runbook essentials

Every adapter has a runbook that answers:

1. **How do I know the regulator is down?** Prometheus alert + status endpoint + last-successful-call metric.
2. **What breaks if it stays down for 1 h? 24 h? 1 week?** Written impact matrix per adapter.
3. **What is the manual fallback?** Documented (portal upload, batch drop, phone call).
4. **How do I replay the queue when it comes back?** Documented replay command with idempotency guarantee.
5. **Who do I call?** Regulator-side contact, escalation chain.

K.6 Observability

Every adapter emits the same telemetry:

Metric	Type	Alert threshold
<code>regulator_calls_total{regulator, status}</code>	counter	Error rate > 5% for 5 min
<code>regulator_latency_seconds{regulator}</code>	histogram	P95 > 10 s for 5 min
<code>regulator_queue_depth{regulator}</code>	gauge	Depth > 10 K for 15 min
<code>regulator_retry_total{regulator}</code>	counter	Retry rate > 10%
<code>regulator_last_success_seconds{regulator}</code>	gauge	> 3600 for payroll-critical
<code>regulator_idempotency_hits_total{regulator}</code>	counter	Investigate spike

Grafana dashboard `regulator-integrations.json` is the single view an on-call engineer opens first when payroll or recruitment misbehaves.

K.7 Deliverable checklist — per integration, before go-live

Every adapter is signed off by **all four** of these people before it enters production:

- ☐ **Solutions Architect** — contract accepted, error taxonomy mapped
- ☐ **Tech Lead** — code review, mock parity verified
- ☐ **Security Engineer** — key rotation drilled, mTLS certificates in production KMS
- ☐ **Operations Lead** — runbook + alert routing + on-call ownership in place

The checklist is what a Steering Committee sees at M6 gate; missing any signature is a blocker.

K.8 Cross-references

- Container-level integration list → [Paper C §Container view](#)
- Regulators named in the PBD → [Paper A §A.3](#)
- PhilSys eKYC as a value-add → [Paper D §13](#)
- Regulator-integration risk items → [Paper J §R8, §R13, §R14](#)
- Migration reliance on PhilSys / DBM PSIPOP → [Paper H §H.2](#)
- Data flowing to regulators, under RA 10173 → [Paper I §I.5](#) and [§I.6](#)